IoT Based Smart Video Surveillance System Using Convolutional Neural Network

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***Abstract*—Video surveillance plays an important role in our times. It is a great help in reducing the crime rate, and it can also help to monitor the status of facilities. The performance of the video surveillance system is limited by human factors such as fatigue, time efficiency, and human resources. It would be beneficial for all if fully automatic video surveillance systems are employed to do the job. The automation of the video surveillance system is still not satisfying regarding many problems such as the accuracy of the detector, bandwidth consumption, storage usage, etc. This scientific paper mainly focuses on a video surveillance system using Convolutional Neural Networks (CNN), IoT and cloud. The system contains multi nods, each node consists of a microprocessor(Raspberry Pi) and a camera, the nodes communicate with each other using client and server architecture. The nodes can detect humans using a pretraining MobileNetv2-SSDLite model and Common Objects in Context(COCO) dataset, the captured video will stream to the main node(only one node will communicate with cloud) in order to stream the video to the cloud. Also, the main node will send an SMS notification to the security team to inform the detection of humans. The security team can check the videos captured using a mobile application or web application. Operating the Object detection model of Deep learning will be required a large amount of the computational power, for instance, the Raspberry Pi with a limited in performance for that reason we used the MobileNetv2-SSDLite model.**

Keywords— IoT, Smart video surveillance, CNN, Deep Learning, Raspberry pi

I. INTRODUCTION

The "Internet of things" (IoT) is becoming a growing subject during the last years. It can be defined as is a system of digital devices, computing, as well as related sensors spread over the globe using the internet which provides the ability to communicate between them for transfer and share information by using unique id which is given to every single device, as Unique Identifiers. [1]

Recently, the amount of video cameras has been increasing over the world, this includes surveillance cameras which are used security, maintenance, healthcare, management, as well as lots of critical purposes. for the 3 reasons Video surveillance sensors, as well as the data of surveillance, are exciting research subjects inside the field of IoT. Firstly, the video surveillance node has transmission bandwidth, which is approximately high, causes obstacles for many of the communications systems. Secondly, video surveillance cameras usually are installed in whatever location such as remote areas. Therefore, the usage of the power to transmit a massive quantity of data such as video sensors is approximately high so that causes higher deployment as well as maintenance costs. Thirdly, the massive information which is collected from different nodes of video surveillance leads to big data. Thus, the filtering of video data, processing along with data mining is additionally a crucial task within IoT[2]. activities, behavior, as Video surveillance systems are extremely important as well as a popular security tool these days. Video surveillance systems used in banks, airports, offices as well as in our homes to keep security.[3]

Each digital video surveillance system contains three modules as shown in Fig. 1:

• Video capture module which consists of a group of cameras.

• Network interface module which is used for processes the video stream as well as sends the video to the central office module.

• Central office module used for process the video that has been received as well as monitors every video channel.[3].

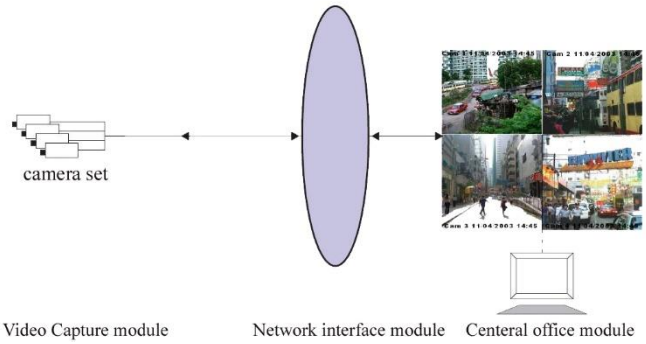


Fig. 1: Video Surveillance System

Deep learning has shown to be a better technique to resolve numerous common complex problems. Several network models have been suggested in deep learning which is centering on many applications. Beginning with Segmentation of the Object to the Generate Image using tag words there are various models being focused on several problems as well as offering an appropriate solution in a more effective manner.

Object detection is a traditional problem in several applications of computer vision. There are A lot of techniques which are actually been created to perform the detection of objects without any human being intervention. Preceding techniques are often dependent on the appearance of the object, more recent methods that are proved to be effective such as utilizing Haar Cascade, as well as particularly Convolution Neural Networks (CNNs). Methods such as

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Faster SDD, R-CNN, YOLO are the example of CNN algorithms that are used for object detection[4][10].

II. Related Work

Video surveillance has been studied from various points of view, the authors in [4] have suggested a system for human detection also tracking on the quadcopter using a Deep Learning algorithm on Raspberry Pi as well as the tracking technique. the authors used for object detection CNN model known as MobileNetv2-SSDLite were utilized for detecting a person. The MobileNetv2-SSDLite model was effective and fast for mobile devices as well as embedded systems. The model MobileNetv2-SSDLite was trained using a Common object in context dataset (COCO) but the author's re-training the model to specializing model for just detected the human and the result after retraining was the mAP is 98.6% using 0.5 for IoU. The advantage of this system is the accuracy of human detection and the disadvantage of this system is there is no monitoring method when humans detected and tracking.

The authors in [5] have suggested Smart Raspberry Pi Surveillance. The system includes raspberry pi employed in surveillance for a deal with the risks of thefts. The system would be to notify the security team when detecting humans within the surveillance location. the system used the SimpleCV library for captures pictures as well as human detection. Notify the security team is performed by sending SMS via GSM Module. The advantage of the system is a low-cost system and the system provided monitoring. the disadvantage of the system is the accuracy of object detection is not efficient.

The authors in [6] have suggested, Identify the objects using Mobile Robots based on ROS. The system contains raspberry pi employed to control the kernel in a mobile robot, Kinect sensor to captures the images and cloud server which is used to detect the object base on the Faster R-CNN model. The Kinect sensor can capture surroundings as well as upload the information to the database in the cloud. Subsequently, the images which are captured are going to be provided to objective identification. The advantage of the system is powerful of detecting since they used the cloud for the detect object, not raspberry pi and the disadvantage of the system is the system send pictures, not videos and the pictures will be sent to the cloud over the internet so there is delay and not a real-time system.

The authors in [7] have suggested a Video Surveillance System using a microprocessor called Rock64, web camera and hardware accelerator from human detection known as Neural Compute Stick (NCS). Due to the higher performance of NCS so it will be used to run the Single Shot Multi-Box Detector (SSD) CNN network in order to detect the human based on streaming video from a web camera. when human has been detected the Rock64 will stream video over IP so the security can check the streaming using a web browser. the advantage of this system is the accuracy of object detection and production time because of the higher performance of NCS which is designed for the Image processing operation. At the same time the disadvantage of the system the web camera not efficient for such a system because of the low quality of the video. Also, the system is much expensive due to the price of NCS and Rock64.

The authors in [8] have suggested to design and implementation of a home security system in order to identify the intruder when no one exists in the home. the system used the Passive infrared sensor(PIR) sensor to detect the intruder based on Raspberry pi. the system will send alarm SMS notification by GSM modem when detecting unauthorized persons using a PIR sensor. the security can monitor the home using IP Camera over the internet connection. Additionally, the system provides the ability to store the videos of the IP camera in cloud storage. the advantage of this system is low cost as well as provide the ability to monitor the home over the internet, but the disadvantage is the PIR sensor very easily interfered by the different heat sources and not work efficiently when the temperature above 35ºC.

III. BACKGROUND KNOWLEDGE

This section will explain the theoretical section of the smart video surveillance system based on IoT.

A. Raspberry Pi

Raspberry Pi is a tiny, cheap, powerful computer board released in 2012 Fig. 2. It operates within the exact way like a PC, needed a keyboard in order to enter the commands, a display screen device, along with electrical power supply. This particular credit card-sized computer with numerous performances and inexpensive. The main components of Raspberry pi 3 as the following below(we will use Raspberry pi 3 model B in the implementation of the system).

- The Raspberry pi 3 models B+ used 1.2 GHz 64-bit quad-core ARM Cortex-A53 CPU and 1GB LPDDR2 SDRAM Ram.
- Raspberry pi 3 B+ contains a slot used for SD Card which usually has the capability to bootable using several Linux distributions and Windows operating systems such as Raspbian, Pidora or Raspbmc, etc. Also, the Raspberry pi supports many programming languages such as Python and Java, which is making build applications easy within Raspberry pi.
- Raspberry pi 3 B+ has 4 ports for USB used to connecting the devices with raspberry pi such as the keyboard, mouse, which makes the raspberry pi as a portable pocket computer.
- Raspberry pi 3 B+ contains the Ethernet port used for connecting the raspberry pi to the network.
- The Raspberry Pi 3 includes an onboard 802.11n Wireless LAN adapter. which is means that they do not need a standalone Wi-Fi dongle. [5].



Fig. 2: Raspberry Pi 3 Board[5]

B. Pi Camera

The Pi Camera component is a camera that can be utilized to capture images as well as videos. Raspberry Pi

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Board contains a Camera Serial Interface(CSI) interface, which is used in order to connect the Pi Camera module to the Raspberry pi board directly. The camera comes with 5 MP Resolution and 1080p @30fps(720p @60fps, 960p @45fps) video recording. The Pi Camera board is very small, approximately 25mm x 20mm x 9mm with weighs 3.5g as shown in fig. 3. The video, as well as image quality, are much better compared to the USB webcam.



Fig. 3: Raspberry Pi Camera

C. Object detection

Object detection is a type of computer vision technique used for locating instances of objects inside the images. the algorithms Object detection algorithms usually used machine learning or deep learning to provide meaningful results.

The humans have the ability to recognize as well as locate objects of interest in just a matter of moments when looking at images or videos so the objective of object detection is replicating this intelligence utilizing a computer. Object detection is not just for classifying or recognizing objects that appear in an image, but also for localizing the objects as well as drawing bounding boxes around these objects[9].

D. MobileNet-V2

MobileNet-v2 has been released as an enhancement of MobileNet-v1. MobileNet-v2 is more powerful as well as more efficient rather than the predecessor.MobileNet-v2 introduces the structure of depthwise separable convolutions into 3 layers rather than 2 layers as in the first version of MobileNet. The last 2 layers in MobileNet-v2 are just like the layers in the first version of MobileNet (MobileNet-v1) which are the depthwise convolution layer as well as the pointwise convolution layer. The first layer of MobileNet-v2 is created as being an expansion layer and the goal is expanding the data of the input layer just before running the depthwise convolution layer. This model came with new features which are inverted residual connections as well as linear bottlenecks. The encodes the inputs and outputs will be done in the bottlenecks block and the inner layer will ensure that the model has the capability to convert from the lower-level concepts like convert pixels to the higher-level descriptors just like image categories. Firstly, to get additional data, the expansion layer will be expanding the input layer due to not a massive amount of information that is extracted when performed the convolutional filters over the low-dimensional feature map. Next, the depthwise separable convolutions are utilized for the expanded layer rather than the normal convolution due to the calculation effectiveness so that is able to reduce 8 to 9 the number of connections.

Nevertheless, distinct from MobileNetv1, the pointwise convolution, within this particular condition, is going to decrease the channels in the preceding layer for packing the data. This concept is named linear bottlenecks which is keeps each channel that belongs to feature maps relatively tiny when compared with different models, as well as low-dimension layers, which would decrease the computational parameter numbers. The remaining connection is usually known as the shortcut among bottleneck layers. those connections assist the stream of the gradients over the network also, will probably be utilized once the input channel volume as well as output channel, will be the same.[4].

E. MobileNet-SSD

Recently, there is actually been a growing interest for developing smaller as well as faster deep neural network architectures suitable for embedded devices. Due to object detection in Real-time and precision which requires huge processing power and on a system with a limited in a performance like embedded devices and mobile, reaching a speed which could be considered as real-time is a difficult task.

So the researches have been built a new model named SSDLite-MobileNetV2 which is low-latency, small as well as low-power models parameterized to satisfy the resource limitations of a diversity of use cases.

object detection algorithm such as Fast R-CNN or Faster R-CNN, that typically produce region proposal areas gotten at the input images which could have an object, and after that generate a prediction according to these regions also, proved to get high accuracy although the speed of the processing is low unlike Single-shot detector techniques such as SSD or YOLO need only to single-pass via the neural network as well as predict in one cycle each of the bounding boxes.

SSDLite was released together with MobileNet-v2 and also, was supposed to be the same SSD model and used in devices that are having a lack of Performance like mobile or embedded systems. All common convolutions in SSD prediction layers are swapped out by the separable depth wise convolutions so the computational cost, as well as parameter count, are reductions. The SSDLite-MobileNetv2 needs 300x300x3 from the image dimension as an input, the MobileNetv2 is utilizing for extract the feature maps, after that detects the objects with several scales by 6 of detection layers. The first layer of detection is connected with the expansion layer of layer 15 from the base model. The others from SSDLite layers are connected to the MobileNetv2 final feature map, as shown in fig. 4

The major purpose of extra layers is usually to independently detect objects during several scales. The first layer of detection which is connected to the expansion layer of the 15 bottleneck layer which is supposed to the small objects will be detected. Since the feature maps are getting smaller because to stride = 2 and the resolution will be smaller as well as the model contain additional details regarding larger objects [4].

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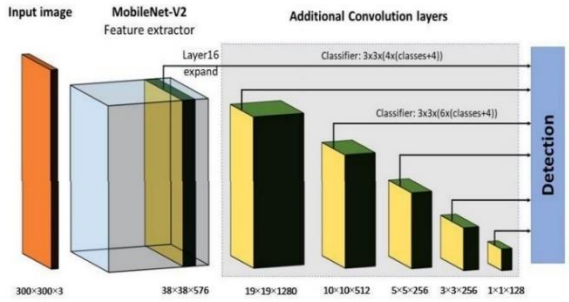


Fig. 4:MobileNetv2-SSDLite[4]

F. TensorFlow

The TensorFlow term came from multidimensional arrays called tensors, which is utilized from the neural networks for various operations. Based on Google, when made a comparison between DistBelief and TensorFlow, TensorFlow has been much faster, more intelligent as well as more flexible also, will be effortlessly adaptable for the newest products as well as areas. TensorFlow was principally developed for the research of deep neural networks and machine learning facilitating. TensorFlow works by sorting by using layers of data (also named nodes) as the learning part. the ability to run on several GPUs as well as CPUs is One of the most effective features of TensorFlow.

G. Public cloud server (Firebase)

Firebase is a technology that enables you to create web applications without server-side programming therefore development will be quicker as well as easier. The Firebase supports web clients, iOS clients, as well as Android clients. The Applications that using Firebase can simply utilize and handle the information, without thinking about how the information will be put in storage and synchronized crosswise with various of the real-time application.

IV. IMPLEMENTATION

A. The Proposed System

In this section, we will describe the video surveillance system which contains multi nodes each node contains raspberry pi and pi camera one of these nodes connect to the cloud which is called the main node and the other nodes called edge node connected to the main node. When the camera on the edge node detects any human motion using CNN, the camera takes video (5 seconds) and streaming it to immediately to the main node using a socket. The main node stream the video to the cloud (using firebase SDK) and send an SMS notification to inform the security that there is human detection so the security can check the videos in mobile application and web application, all that should be done in real-time.

One important component of deep learning is the dataset. An efficient dataset is going to contribute to a model with great precision as well as recall. In the field of object detection in the images, there are several household names widely used and indicated by researchers such as ImageNet,

Pascal, COCO, YOLO. In this paper, We will use the COCO dataset, due to its distinct feature and labeled objects

Algorithm 1 Human detection and streaming

```
while true do
    frame ← new frame
    if (node is detected, person)
        if(detected node is an edge)
            call: MainNodeProcessing(video)
        else if (detected node is main)
            call: MainNodeProcessing(video)
        end if
    end if
end while
Function MainNodeProcessing(video)
    stream video to the firebase cloud
send SMS alert to the security team
End function
```

B. Architecture

For the implementation of, the following items are required :

• Raspberry Pi 3- contain Wi-Fi-connectivity.

• Pi Camera.

• Object detection using a pretrained MobileNetv2-SSDLite model and uses the MS-COCO dataset. The most challenges when deploying a Convolutional Neural Network (CNN) in an embedded device with low graphics capacity is the time of processing of the images on the device. As a result, it is required to use CNN model which can capable with this limitation of raspberry pi which is MobileNetv2-SSDLite.

• SMS Web Service (Twilio).

• Cloud (Firebase service).

• Android phone as well as a web application to check the videos.

The block diagram as shown in fig. 5 will be described each component of the system.

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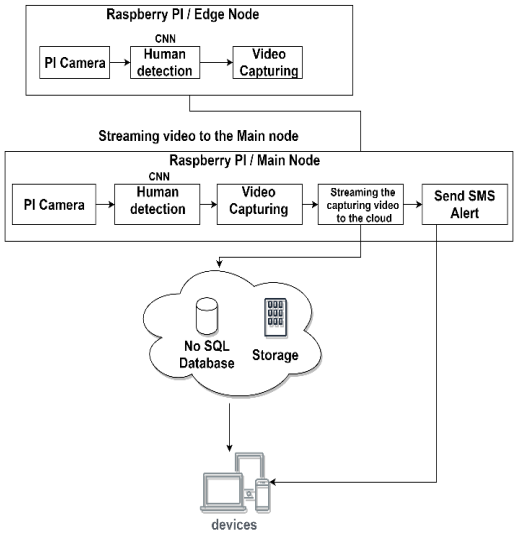


Fig. 5: System Block diagram

C. Raspberry pi /Edge node

These nodes consist of Raspberry pi and PI camera. Using for detected humans based on the SSDLite-MobileNet-V2 CNN model. The camera will capture a 5-second video using OpenCV library when humans detected and streaming the video to the main node as shown in the following algorithm 2

Algorithm 2 Raspberry pi /Edge node
1: while true do
2: frame ⟵ new frame
3: if (edge node is detected, person)
4: Stream the video to the main node
5: end if
6: end while

D. Raspberry pi /Main node

This node contains Raspberry pi, PI camera, and Internet connection. the main node has 2 functions, the first is to detect humans Depending on the SSDLite-MobileNet-V2 CNN model. The camera will capture a 5-second video using the OpenCV library when humans detected. The second function, At the same time this node has the ability to receive the video from edges node and in the 2 functions of the main node the video will be streamed to the cloud-based on the internet connection as shown in the following algorithm 3

Algorithm 3 Raspberry pi /Main node
1: while true do
2: frame ⟵ new frame
3: if (the Main node is detected, person)
4: call: MainNodeProcessing(video)
5: end if
6: if (the Main node is received video)
7: call: MainNodeProcessing(video)
8: end if
9: end while
10:Function MainNodeProcessing(video)
11: stream video to the firebase cloud
12: send SMS alert to the security team
13: End function

V. RESULTS

The evaluation metrics used to evaluate performance for video surveillance systems are Mean Average Precision (mAP) and Predication time. the mAP is depended on other metrics such as Precision, Recall, PR curves and Intersection Over Union (IoU). The Mean Average Precision (mAP) for human detection for the first scenario is 96.1% based on The Precision-Recall (PR) curves as shown
in below fig. 8 are drawn to highlight the relationship between precision and recall at different threshold values (different from IoU threshold values). The threshold IOU is 0.5 which is chosen base on testing. We used the MobileNetv2-SSDLite model for human detection since The Raspberry Pi has a weak processor, so we need to use a model that takes less processing power, which is the fastest model

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available for the embedded system and mobile. If we compare the accuracy of human detection with

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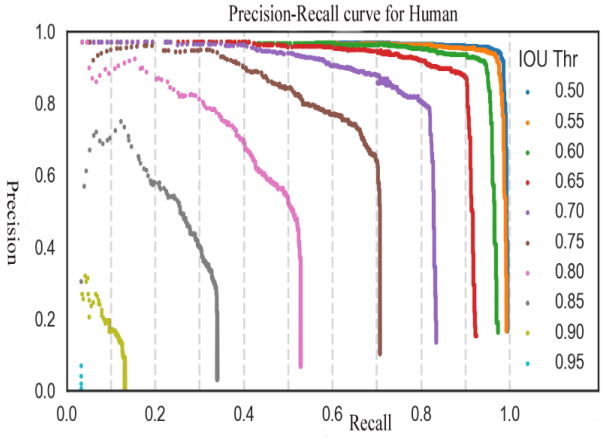


Fig. 6: mean average precision

The average of prediction time for human detection approximately is 1.72 seconds, which is the time that takes the MobileNetv2-SSDLite model for prediction the human and dependent on raspberry pi hardware resource.

VI. Conclusion

The research work that has been performed in this scientific paper has been primarily focused on design as well as develop an effective and powerful video surveillance system that will help to reduce and stop theft. in spite of a significant amount of research that has been performed previously to resolve these kinds of security problems, however, it remains to be challenging because of increased complexity as well as several theft actions are happening every day.

The system will be used the MobileNetv2-SSDLite model to detect a human due to efficient performance when used in mobile devices as well as embedded systems. Also, This model will be used pre-trained on Common object in context dataset (COCO) because of a huge amount of the images. The system will capture video once the motions exceed a specific threshold which is pre-set in the system about 5 seconds and send it to the main node so the main node will be sent the video to the cloud as well as send an alarm to the security team. therefore, this reduces the amount of data that usually needs to be reviewed. Additionally, it is going to help save data space by not recording videos that usually does not include the object of interest. Such as system will be usable in the office as well as home. Such alarm systems are massively wanted for security functions, and therefore the specified system could be proved effective as well as useful in view of the above-mentioned features.

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